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Pulsed capacitively coupled plasma processes

Abstract

A method of plasma processing includes cyclically performing a cycle including the steps of performing a glow phase and performing an afterglow phase. The glow phase includes providing a first SP pulse comprising a first SP power level for a first duration to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber. The first SP pulse terminates at the end of the glow phase. The afterglow phase is performed after the glow phase and includes providing a BP pulse train to a BP electrode coupled to a target substrate within the plasma processing chamber in an afterglow of the capacitively coupled plasma for a second duration between about 10 μ s and about 100 μ s. The BP pulse train includes a plurality of BP spikes. Each of the plurality of BP spikes is a DC pulse that has a first BP power level.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/001,327, filed on Aug. 24, 2020, which application is hereby incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to plasma processing, and, in particular embodiments, to systems and methods for plasma processing using capacitively coupled plasma and pulsed power.

BACKGROUND

(2) Device fabrication within microelectronic workpieces may involve a series of manufacturing techniques including formation, patterning, and removal of a number of layers of material on a substrate. Many such techniques require high aspect ratios such as high aspect ratio etches (e.g. memory etches for NAND fabrication), high aspect ratio contact (HARC) etches (e.g. for logic contacts), as well as other back end of line (BEOL) etches.

(3) Modern semiconductor device fabrication processes seek to push high aspect ratio processes for even greater limits. For example, attempting to meet strict tilt, aspect ratio, and ellipticity requirements, methods utilizing low temperature, tailored waveforms, and/or synchronous power (i.e. source power and bias power on simultaneously) combined with very high source power and bias power have been employed.

(4) Capacitively coupled plasma (CCP) systems may be used for high aspect ratio processes. Higher power and voltage, additional frequencies, and lower processing temperatures may all be desirable to improve process control, but add considerable complexity to the CCP system. Therefore, more agile hardware enables precision control of flux, energy, and chemistry during plasma processing without employing while achieving lower complexity may be desirable.

SUMMARY

(5) In accordance with an embodiment of the invention, a method of plasma etching includes performing a first on phase including applying a source power (SP) pulse to an SP electrode to generate plasma in a plasma processing chamber, performing a second on phase after the first on phase, performing a corner etch phase after the second on phase, and performing a by-product management phase after the corner etch phase. The SP pulse includes a first SP power level and terminates at the end of the first on phase. The second on phase includes applying a first bias power (BP) pulse to a BP electrode coupled to a target substrate within the plasma processing chamber. The first BP pulse includes a first BP power level and accelerates ions of the plasma toward to target substrate to etch a recess in an etchable material of the target substrate. The corner etch phase includes applying a BP spike including a second BP power level greater than the first BP power level. The duration of the BP spike is less than the duration of the first BP pulse. The by-product management phase includes applying source power to the SP electrode at a second SP power level that is less than the first SP power level and applying bias power to the BP electrode at a third BP power level that is less than the first BP power level.

(6) In accordance with another embodiment of the invention, a method of plasma processing includes performing a glow phase including providing a first SP pulse including a first SP power level to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber and providing a first BP pulse including a first BP power level to a BP electrode coupled to a target substrate within the plasma processing chamber. The first SP pulse terminates at the end of the glow phase. The method further includes performing an afterglow phase after the glow phase. The afterglow phase includes providing a second BP pulse to the BP electrode in an afterglow of the capacitively coupled plasma. The second BP pulse includes a second BP power level that is less than the first BP power level.

(7) In accordance with still another embodiment of the invention, a method of plasma processing includes cyclically performing the following steps: concurrently applying SP including a first SP power level to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber, and BP including a first BP power level to a BP electrode coupled to a target substrate within the plasma processing chamber, reducing the SP from the first SP power level to a second SP power level, reducing the BP from the first BP power level to a second BP power level, and reducing the BP from the second BP power level to a third BP power level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIGS. 1A, 1B, and 1C illustrate an example plasma processing method in accordance with an embodiment of the invention, where FIG. 1A illustrates a schematic timing diagram of the plasma processing method, FIG. 1B illustrates a corresponding qualitative graph, and FIG. 1C illustrates a corresponding target substrate;

(3) FIG. 2 illustrates a schematic diagram of a target substrate during an example plasma process utilizing balanced control of plasma parameters in accordance with an embodiment of the invention;

(4) FIG. 3 illustrates a schematic diagram of another target substrate during an example plasma process utilizing imbalanced control of plasma parameters in accordance with an embodiment of the invention;

(5) FIGS. 4A, 4B, and 4C illustrate another example plasma processing method in accordance with an embodiment of the invention, where FIG. 4A illustrates a schematic timing diagram of the plasma processing method, FIG. 4B illustrates a corresponding qualitative graph, and FIG. 4C illustrates a corresponding target substrate;

(6) FIG. 5 illustrates a schematic timing diagram of yet another example plasma processing method in accordance with an embodiment of the invention;

(7) FIG. 6 illustrates a schematic timing diagram of still another example plasma processing method in accordance with an embodiment of the invention;

(8) FIGS. 7A, 7B, and 7C illustrate still yet another example plasma processing method in accordance with an embodiment of the invention, where FIG. 7A illustrates a schematic timing diagram of the plasma processing method, FIG. 7B illustrates a corresponding qualitative graph, and FIG. 7C illustrates a corresponding target substrate;

(9) FIG. 8 illustrates a block diagram of an example capacitively coupled plasma system in accordance with an embodiment of the invention;

(10) FIG. 9 illustrates an example method of plasma processing in accordance with an embodiment of the invention; and

(11) FIG. 10 illustrates another example method of plasma processing in accordance with an embodiment of the invention.

(12) Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale. The edges of features drawn in the figures do not necessarily indicate the termination of the extent of the feature.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(13) The making and using of various embodiments are discussed in detail below. It should be appreciated, however, that the various embodiments described herein are applicable in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use various embodiments, and should not be construed in a limited scope.

(14) As the capabilities of high aspect plasma processes are expanded requirements such as tilt mitigation, aspect ratio-dependent etch mitigation, and ellipticity mitigation become more strict and harder to achieve (e.g. add very high power). Several techniques have been employed in attempts to meet these requirements. For example, low temperature plasma processing and tailored waveforms (e.g. combining harmonic frequencies to form various waveforms such as square wave approximations) have been used. However, these techniques are complex solutions which may be expensive and difficult to implement successfully. Further, complexity may also reduce flexibility as a result of the large number of interdependent pieces.

(15) CCP plasma is desirable for high aspect ratio plasma processes because CCP plasma is of medium density and can couple very high voltage biases. For example, medium density plasma may also result in low degrees of etch precursor dissociation which may be necessary for selectivity and rate. CCP plasma may be particularly useful for dielectric etches such as silicon oxide (SiO_2), silicon nitride (Si_3N_4), and low- κ dielectric etches. For example, CCP plasma may be used in dielectric etches for memory structures (e.g. NAND layers).

(16) Modifications to the application of power to the plasma system may also be used such as pulsed application of source power (SP) and bias power (BP). The source power pulse may include high frequency (HF) radio frequency (RF) power while the bias power pulses may include low frequency (LF) RF power with a direct current (DC) offset or simply be direct current (DC) power. Other RF frequency ranges may of course be used such as very high frequency (VHF), medium frequency (MF), very low frequency (VLF), and others.

(17) Conventional pulsing methods use synchronous pulsing involving simultaneous or predominantly overlapping applications of SP pulses and BP pulses. Yet conventional synchronous pulsing has the drawback severely reduced effectiveness of the bias power to increase the voltage when the source power is on. As a result, higher power must be employed for BP pulses in an attempt to increase the voltage in the presence of the SP pulses.

(18) Yet blind application of high power and synchronous pulsing may be especially problematic for CCP systems. For example, ion energy may be relatively low considering the magnitude of applied power. Additionally, ion temperature may be high resulting in thermally dominated ion motion. The dissociation rate in the plasma may also be relatively low which may result in lack of chemistry control, minimal ion energy/angle control, and increased deposition during the off phase (e.g. polymer may be dumped to walls and the wafer). Process space may be (marginally) increased for synchronous pulsing technology by utilizing enormous power (e.g. tens of kW for both source and bias power) but this is suboptimal for at least the aforementioned reasons in addition to being energy inefficient.

(19) Additionally, trade-offs exist when utilizing synchronous pulsing. Synchronous pulsing techniques may use HF RF power for the SP pulses and LF RF power for the BP pulses. Large HF power is needed to create desired flux, but increased flux suppresses ion energy. Conversely, large LF power is needed for high ion energy and verticality. As the HF power increases, the plasma density increases making it harder to generate high energy ions at reasonable power resulting in

very large power supplies for the BP pulses (e.g. 20 kW, 40 kW, etc.). But as the voltage is increased by larger LF power, sheath thickness is also increased, reducing the volume of the plasma and requiring more HF power for the SP pulses to maintain the plasma. The plasma generation zone is squeezed due to the larger sheath which then decreases the ion density and the ion flux.

(20) The inherent coupling of source and bias power during synchronous application of SP pulses and BP pulses utilized in conventional pulsing methods prevents desirable decoupling of ion energy and the plasma chemistry. The result is reduced control over plasma parameters and diminished precision during plasma processing. The embodiment systems and methods of plasma processing described herein overcome these shortcomings using staggered multiphased pulsing schemes to independently control the ion to radical ratio and maximize the ability to extract voltage.

(21) The plasma processing methods plasmas described herein may advantageously afford enhanced and independent control over various plasma parameters. For example, increased ion flux, radical flux, and the ratio between the two may be advantageously achieved. A further possible benefit of the embodiment plasma processing methods is increased control over ion energy and radical energy. In particular, desirable maximization of ion energy and ion verticality can be advantageously obtained by included antiphased the source and bias powers.

(22) Another possible benefit of the embodiment plasma processing methods is enhanced and independent control over plasma chemistry. For instance, chemical ratios such as the fluorine (F) to carbon (C) ratio may be beneficially controlled using additional pulse phases as described herein. Enhanced chemistry control using different phases may advantageously provide appropriate degrees of polymer buildup throughout the plasma process to prevent undesirable effects such as clogging (e.g. at the opening of a small via) or cue-tipping (bowing of the mask), and mask erosion.

(23) The plasma processing methods plasmas described herein may also advantageously enable improved process margin over conventional techniques. For example, process margin may be improved for HARC etches (e.g. for logic contacts), NAND memory etches, and other BEOL etch processes as well as other high aspect ratio processes. Additionally, the embodiment plasma processing methods may use CCP systems and methods which advantageously maintain the benefits of CCP for HARC, NAND, BEOL, and others while adding various benefits of antiphased pulsing techniques. Such combined use of capacitive coupling and antiphased pulsing may advantageously generate high power and high ion energies improving aspect ratio (e.g. up to 100, and higher).

(24) Embodiments provided below describe various systems and methods for plasma processing, and in particular, systems and methods for plasma processing that include applying pulsed source power and bias power to CCP plasmas. The following description describes the embodiments. FIGS. 1A, 1B, and 1C are used to describe an embodiment plasma processing method. Two target substrates during embodiment plasma processes are described using FIGS. 2 and 3. Another embodiment plasma processing method is described using FIGS. 4A, 4b, and 4C. FIGS. 5 and 6 are used to describe another pair of embodiment plasma processing methods. FIGS. 7A, 7B, and 7C are used to describe another embodiment plasma processing method. An embodiment capacitively coupled plasma processing system is described using FIG. 8. Three embodiment methods of plasma processing are described using FIGS. 9 and 10.

(25) FIGS. 1A, 1B, and 1C illustrate an example plasma processing method in accordance with an embodiment of the invention, where FIG. 1A illustrates a schematic timing diagram of the plasma processing method, FIG. 1B illustrates a corresponding qualitative graph, and FIG. 1C illustrates a corresponding target substrate.

(26) Referring to FIGS. 1A, and 1B, a schematic timing diagram **100** includes source power SP and bias power BP provided as pulses to generate reactive species and energetic ions over various phases of a cycle **150**. For example, the cycle **150** may be an advanced pulsing technique (APT) applied to a CCP system. The cycle **150** includes various phases characterized by power and duration of both the SP and the BP (e.g. 4 phases as shown). In various embodiments, the cycle **150**

is repeatedly performed (e.g. cyclically). For example, the cycle **150** may be performed many times (e.g. $\gg 1$), the exact number of times depending on the specific objectives of a chosen plasma process.

(27) The cycle **150** includes a first on phase **110** during which a first SP pulse **112** is applied with power $P_{\text{sub.S1}} > 0$. A first BP pulse **114** with power $P_{\text{sub.B1}}$ may also be applied during the first on phase **110**. The first on phase **110** is defined by a duration $t_{\text{sub.1}}$ equal to the pulse duration of the first SP pulse **112**. The first BP pulse **114** may or may not extend to the end of the first on phase **110** as illustrated by an optional extended BP pulse **113**.

(28) The duration $t_{\text{sub.1}}$ of the first on phase **110** is greater than about 10 μs . In some embodiments, $t_{\text{sub.1}}$ is between about 10 μs and about 100 μs . In one embodiment, duration $t_{\text{sub.1}}$ of the first on phase **110** is about 20 μs .

(29) During the first on phase **110**, there may be a higher degree of plasma and radical generation in conjunction with a lower degree of imparted ion energy and etching. The first on phase **110** may be referred to as an SP phase (e.g. since higher power SP dominates the effects on the plasma), a plasma generation phase or a glow phase due to the characteristic plasma and radical generation.

(30) The first on phase **110** affects various plasma parameters as illustrated in qualitative graph **102** of FIG. **1B** including an ion flux curve **105**, a radical flux curve **106**, an ion energy curve **107**, and a by-product flux curve **108**. During the first on phase **110**, the ion flux $\Gamma_{\text{sub.i}}$ and the radical flux $\Gamma_{\text{sub.r}}$ are both high due to the application of SP in the first SP pulse **112**. When BP is included in the first on phase **110**, some ion energy $\epsilon_{\text{sub.i}}$ may be generated. The first BP pulse **114** may be included in order to impart some $\epsilon_{\text{sub.i}}$ (e.g. for thickness control). However, $\epsilon_{\text{sub.i}}$ is low due to the reduced effectiveness of applying BP while also applying SP. Low $\epsilon_{\text{sub.i}}$ also may result in minimal etching and low by-product flux $\Gamma_{\text{sub.bp}}$.

(31) During the first on phase **110**, there may be high plasma current due to increased plasma density during plasma generation. $P_{\text{sub.B1}}$ may be relatively high in order to achieve the high bias voltage that may be required because of the high plasma current. However, in very electronegative cases, the plasma density may be low during the first on phase **110** and relatively high $P_{\text{sub.B1}}$ may not be necessary.

(32) During the source power phase (first on phase **110**) SP is high and a plasma glow is maintained. The SP is reduced after the source power phase and the generated plasma enters an afterglow state. The afterglow may be useful because the plasma density is dropping and temperatures (ion and electron) are dropping advantageously allowing the ion angle to be narrowed and ion energy to be increased.

(33) Advantageously, $\epsilon_{\text{sub.i}}$ may be maximized and ion verticality optimized when the ion density drops and the temperature drops in the afterglow of the SP phase. As a result, a second on phase **120** follows the first on phase **110** during which a second BP pulse **124** with duration $t_{\text{sub.2}}$ and power $P_{\text{sub.B2}} > 0$ is applied. During the second on phase **120** the SP is lower ($P_{\text{sub.S2}} < P_{\text{sub.S1}}$) and may be zero. The inclusion of a second SP pulse **122** in the second on phase **120** may increase the absolute value of $P_{\text{sub.B2}}$ to compensate for its reduced effect of the bias voltage in the presence of SP.

(34) The duration $t_{\text{sub.2}}$ of the second on phase **120** is greater than about 10 μs . In some embodiments, $t_{\text{sub.2}}$ is between about 10 μs and about 100 μs . In one embodiment, duration $t_{\text{sub.2}}$ of the first on phase **120** is about 40 μs .

(35) As shown in qualitative graph **102**, the ion energy $\epsilon_{\text{sub.i}}$ increases in the second on phase **120**. Because of the increase in $\epsilon_{\text{sub.i}}$ from applied BP to the plasma afterglow, the second on phase **120** may be referred to as a BP phase, an afterglow phase, or a main etch phase (since increased $\epsilon_{\text{sub.i}}$ facilitates etching). The by-product flux $\Gamma_{\text{sub.bp}}$ may sharply increase from by-product formation during etching. Ions are no longer being generated in the afterglow and are depleted from the plasma by the BP so the ion flux $\Gamma_{\text{sub.i}}$ may decrease while the radical flux $\Gamma_{\text{sub.r}}$ may remain relatively constant.

(36) As the bias voltage increases, the ion energy ($\epsilon_{\text{sub.i}}$) increases and the ion angular distribution narrows. Ion energy control and angle control during the second on phase **120** allows it to be used as a main etch phase. When included and as shown, the first BP pulse **114** may end before the end of the first SP pulse **112** (i.e. be reduced to the level of the second BP pulse **124** prior to the reduction or removal of SP power in the second on phase **120**). This may advantageously prevent or reduce an undesirable voltage spike upon entering the second on phase **120**.

(37) There may also be a delay between the end of the first on phase **110** (end of the first SP pulse **112**) and the beginning of the second on phase **120** (beginning of the second BP pulse **124**). Both the ion to radical ratio and $\epsilon_{\text{sub.i}}$ may also be a function of the delay between the phases as well as the optional source power during the second BP pulse **124**. If a first BP pulse **114** is included in the first on phase **110**, then $P_{\text{sub.B2}}$ may be less than $P_{\text{sub.B1}}$ since the SP during the second on phase **120** is lower and may be omitted entirely.

(38) Several schemes involving pulsed application of SP and BP pulses may be implemented. For example, the SP may be kept constant (at a power level suitable to perpetually sustain the plasma) while only the BP is pulsed (bias pulsing). In this scenario, flux is always high due to a continuous plasma glow. However, ion energy is low when the BP is on and even lower when the BP is off (i.e. at the plasma potential). That is, the ion energy cannot be effectively increased due to the SP being on because of the high plasma density.

(39) As mentioned previously, SP pulses and BP pulses may also be applied synchronously (synch pulsing). In this implementation, there is a chemistry change between on and off phases, but no independent chemistry control. Further, there is weak ion energy and angle control because the SP is always on at the same time as the BP. Ion energy is again low in the on phase and then lower in the off phase (i.e. thermal). A large amount of polymer deposition may occur in the off phase.

(40) Another pulsing scheme is to maintain the BP at a constant level and pulse the SP (source pulsing). In contrast to bias pulsing and synch pulsing, source pulsing includes phases where only the BP is on and therefore may utilize an afterglow phase. However, the BP has severely reduced effectiveness when the SP is on since the BP is held constant. Further, when both the SP and the BP are on the plasma generation zone may be squeezed (i.e. diminished) due to a larger sheath reducing the ion concentration and requiring more power.

(41) The SP pulses and the BP pulses may also be pulsed independently from one another (asynchronous pulsing). Asynchronous pulsing may include partially overlapping SP and BP pulses or fully nonoverlapping SP and BP pulses. Some SP and BP pulses may be synchronous while others are asynchronous in an asynchronous pulsing implementation. The power of the SP and BP may also be different at different times. For this reason asynchronous pulsing implementations may advantageously provide the flexibility to decouple and enhance control of various plasma parameters during plasma processes.

(42) Still referring to FIGS. **1A** and **1B**, the cycle **150** further includes an optional third phase **130** during which a third BP pulse **134** is applied for a duration $t_{\text{sub.3}}$ with power $P_{\text{sub.B3}} > P_{\text{sub.B2}}$. For example, the optional third phase **130** may be a shorter phase ($t_{\text{sub.3}} < t_{\text{sub.2}}$ and/or $t_{\text{sub.1}}$) and may include a bias power $P_{\text{sub.B3}}$ that is very high (e.g. as high as possible). Because of the short duration of $t_{\text{sub.3}}$ and large magnitude of $P_{\text{sub.B3}}$, the third BP pulse **134** may be referred to as a BP spike. For this reason, the optional third phase **130** may be referred to as a BP spike phase.

(43) The duration $t_{\text{sub.3}}$ of the optional third phase **130** is greater than about 10 μs . In some embodiments, $t_{\text{sub.3}}$ is between about 10 μs and about 30 μs . In one embodiment, duration $t_{\text{sub.3}}$ of the optional third phase **130** is about 20 μs .

(44) The purpose of very high $P_{\text{sub.B3}}$ may be to create a voltage spike (and ion energy spike) at the end of the BP phase. That is, the optional third phase **130** may be viewed as part of the BP phase occurring for a brief time $t_{\text{sub.3}}$ relative to $t_{\text{sub.2}}$ at the end of the second on phase **120**. During the optional third phase **130**, the SP is removed or further reduced (including optional an optional third SP pulse **132** as shown).

(45) During the optional third phase **130**, $\Gamma_{\text{sub.i}}$ spikes while $\Gamma_{\text{sub.i}}$ continues to decrease due to the lack of plasma generation and further depletion of the ions from the plasma. A corresponding spike in $\Gamma_{\text{sub.bp}}$ may also occur due to etching from energetic ions. The size of this $\Gamma_{\text{sub.bp}}$ spike may smaller than the spike in the second on phase **120** since there are fewer ions and fewer energetically favorable regions than at the start of the BP phase.

(46) An optional fourth phase **140** may also be included in the cycle **150**. During the optional fourth phase **140**, both the SP and the BP may be reduced or removed entirely ($P_{\text{sub.S4}} < P_{\text{sub.S1}}$, $P_{\text{sub.B4}} < P_{\text{sub.B2}}$) for a duration $t_{\text{sub.4}}$. For example, a fourth BP pulse **144** may be included or omitted. Similarly, a fourth SP pulse **142** may be included. The power of the third SP pulse **132** may be similar to the power of the fourth SP pulse **142**. That is, if one is omitted both may be omitted and if one is included, both may have the same power. However, this is not a requirement.

(47) The duration $t_{\text{sub.4}}$ of the optional fourth phase **140** is greater than about 100 μs . In some embodiments, $t_{\text{sub.4}}$ is between about 100 μs and about 2 ms. In one embodiment, duration $t_{\text{sub.4}}$ of the optional fourth phase **140** is about 200 μs .

(48) The optional fourth phase **140** may provide a delay between the BP phase and the SP phase of the next cycle to allow by-products to be removed from a plasma processing chamber (i.e. pumped out). Since by-products may be removed and controlled by reduced or removed power, the optional fourth phase **140** may be referred to as an off phase or a by-product management phase.

(49) In some cases by-product buildup may still occur at undesirable levels during the optional fourth phase **140** in the absence of any power. Therefore, as shown, just enough SP and/or BP may be applied during the optional fourth phase **140** to control buildup. This may result in slight increases in $\Gamma_{\text{sub.i}}$, $\epsilon_{\text{sub.i}}$, and $\Gamma_{\text{sub.bp}}$ during the optional fourth phase **140** as indicated by the alternate off-phase curves **109** in FIG. **1B**.

(50) Referring now to FIG. **1C**, a target substrate **304** during various stages of processing corresponding to the phases of the plasma processing method of FIGS. **1A** and **1B** is shown. In this specific example, the plasma processing method is applied to a high aspect ratio etching process. The target substrate **104** includes an etchable material **160**. In some embodiments, the etchable material **160** comprises silicon. In one embodiment the etchable material **160** is bulk silicon. In various embodiments, the etchable material **160** is a dielectric and is SiO_2 in one embodiment. Alternatively, the etchable material may be Si_3N_4 , a high- κ dielectric, and others. Further, the etchable material **160** may also be a multilayer structure (e.g. alternating layers of $\text{SiO}_2/\text{Si}_3\text{N}_4$, or of Si/SiO_2 , etc.).

(51) A mask **161** is disposed over the etchable material **160**. The mask **161** may protect the etchable material **160** during the etching process. The target substrate **104** is shown here at an intermediate step where a recess **162** has already been formed through an opening in the mask **161**. For example, the recess **162** may be a hole (substantially 1D structure) or a trench (substantially 2D structure).

(52) During the first on phase **120**, polymer may be formed at exposed surfaces of the mask **161** and the etchable material **160**. The polymer may form mainly at top surfaces of the mask and build up at the bottom of the recess **162** in the corners (illustrated as corner buildup **163**). Polymer buildup may advantageously protect mask surfaces and sidewalls as well as form a mixing layer **165** with the etchable material **160** that facilitates etching. For instance, the mixing layer **165** may include species from both the etchable material **160** and the etchant (e.g. SiO_2 and CF_2).

(53) As a specific example, the high aspect ratio etching process may be an oxide etch that uses a fluorocarbon (C_xF_y) etchant. At a high level, chemistry control for such a process may be thought of as controlling the ratio of fluorine to fluorocarbon species at the target substrate **104**. For example, CF_2 species (radicals and ions) may have a particularly high representation among the various generated fluorocarbons. Therefore, chemistry control can be conceptually thought of as controlling ratio $\text{F}:\text{CF}_2$ at various phases of the oxide etch. In the first on phase **120**, C_xF_y species become dissociated in the plasma generating F, CF_2 , and

CF.sub.2.sup.+ (among potentially many other species) which causes F:CF.sub.2 to increase.

(54) After the first on phase **110**, applied BP in the afterglow caused by the reduction or removal of SP during the second on phase **120** facilitates etching of the oxide in the recess **162**. Ions (e.g. CF.sub.2.sup.+) generated in the first on phase **110** are given energy by the bias voltage from the BP and may bombard the mixing layer **165** to etch the bottom of the recess **162** to a cycle etch depth **166** (e.g. depth per BP pulse in the second on phase **120**).

(55) In contrast to the previous phase, F:CF.sub.2 decreases in the second on phase **120** (e.g. due to recombination). Increased duration (t.sub.2) may advantageously allow greater leverage of this chemistry change. For instance, more carbon may be desirable to protect the mask during the main etch since the ions are more energetic.

(56) By-products **167** that are generated during the second on phase **120** may also deposit on the top and sidewalls of the mask **161** and sidewalls of the recess **162**. This by-product buildup may also provide protection through passivation. However, the by-products **167** may also deposit in the corners and bottom of the recess **162**. As shown, the by-products **167** may further exacerbate the effects of the corner buildup **163** that begin in the first on phase **110**.

(57) HF power may be used for the SP pulses while LF power may be used for the BP pulses during the cycle **150**. In one embodiment, the SP pulses comprise RF power with frequency greater than about 13.56 MHz and about 40 MHz in one embodiment. In other embodiments, the HF power may be 27 MHz, 40 MHz, 60 MHz, 100 MHz, 200 MHz, and higher. The lower the frequency of the LF (BP) power, the greater the coupling resulting in higher ion energy E. For example, sufficiently low frequency BP so as to be at a bimodal frequency may be desirable. In some embodiments the BP pulses comprise RF power with frequency less than about 13 MHz and about 500 kHz in one embodiment. For example, the LF power may be in the range of about 100 kHz to about 13 MHz. In other embodiments, the BP pulses are DC pulses.

(58) When included, the optional third phase **130** may serve as a corner etch phase by generating an energy spike that narrows the ion angular distribution for interior corner etch **168**. The lower SP combined with very high BP may advantageously increase the effectiveness of the BP pulse increasing the voltage and generating an ion energy spike. Since the majority of ions in the afterglow have been depleted during the second on phase **120**, there the ion flux F_i is lower thereby beneficially reducing the risk of sputtering while still providing highly energetic ions to the bottom of the recess **162**.

(59) The narrow angular distribution and high energy of the ions in the optional third phase **130** may advantageously deliver desired energy to clean up the corners. This widens the undesirable 'V' shape at the bottom of the recess **162** as shown schematically in FIG. 1C. Due to the low flux, short duration, and the lack of a mixing layer at the bottom of the recess **162**, the etch depth may not be significantly affected by the optional third phase **130**. Since the plasma process may be performed cyclically, the optional third phase **130** may be used to control buildup of films.

(60) The optional fourth phase **140** may be included for mask and sidewall cleanup **169**. For example, by-products that are less volatile than desired (e.g. residual hard mask material, etc.) may be pumped out during the optional fourth phase **140**. However, in some applications (e.g. dielectric etches) the optional fourth phase **140** may not be used because F may sufficiently volatilize Si products resulting in negligible by-product buildup per cycle.

(61) The optional fourth phase **140** may advantageously reduce clogging in future cycles of the plasma process. In some cases, some SP and/or BP may be required to release by-products from the mask and sidewall surfaces to allow their removal from the system. The duration of the optional fourth phase **140** may be on the order of the residence time of the by-products.

(62) In general, the duty cycle (e.g. the percent on time for each pulse in a given cycle) may vary widely depending on application specifics. For example, each of the phases may be anywhere in the range of about 3% to about 90%. Further, the specific timing of each phase may also depend on the pulse frequency. For example, very long phases may reduce the pulse frequency.

(63) FIG. 2 illustrates a schematic diagram of a target substrate during an example plasma process utilizing balanced control of plasma parameters in accordance with an embodiment of the invention. The target substrate of FIG. 2 may be a specific implementation of other target substrates described herein such as the target substrate of FIG. 1C, for example. Similarly labeled elements may be as previously described.

(64) Referring to FIG. 2, a target substrate **204** includes an etchable material **260**, a mask **261**, and a recess **262**. It should be noted that here and in the following a convention has been adopted for brevity and clarity wherein elements adhering to the pattern “_10” may be related implementations of a plasma processing chamber in various embodiments. For example, the etchable material **260** may be similar to the etchable material **160** except as otherwise stated. An analogous convention has also been adopted for other elements as made clear by the use of similar terms in conjunction with the aforementioned three-digit numbering system.

(65) During a plasma process such as any of the embodiment plasma processes described herein, plasma parameters may be controlled so as to achieve desirable results. The specific example illustrated schematically by target substrate **204** is of a plasma etching process (e.g. high aspect ratio) in which a fluorocarbon (FC) is used as an etchant. This process may be useful for HARC and NAND applications and others. The FC etchant may be any suitable FC species having the chemical formula $C_{x_1}F_{y_1}$. For example, the FC etchant may be CF_4 , C_4F_8 , C_5F_8 , C_2F_6 , C_4F_6 , C_5F_6 , and others. The FC etchant may also be a hydrofluorocarbon (e.g. CHF_3 , CH_3F) and/or in combination with O_2 , H_2 , NF_3 , SF_6 , Ar, He, etc.

(66) Three primary categories of species may be delivered to the target substrate **204** as shown: FC radicals **71**, F radicals **72**, and ions **73**. The FC radicals **71** may include any radicals comprising both F and C ($F_{x_1}C_{y_1}$). In one embodiment the FC radicals **71** comprise CF_2 radicals (as shown). In some instances many different FC radical species may be delivered to the majority of the FC radicals **71** being CF_2 . The F radicals **72** are electrically neutral F atoms.

(67) A large variety of ions **73** may be generated in the plasma during the plasma etching process. In various embodiments, the ions **73** comprise FC ions. In some embodiments, the ions **73** comprise $C_{x_1}F_{y_1}^{+}$ ions and comprise CF_2^{+} ions in one embodiment. For example, as above, although many species of FC ions may exist, the ions **73** may predominantly include CF_2^{+} ions. The ions **73** may also comprise F ions among others (e.g. ions from inert species such as Ar^{+}).

(68) During the plasma etching process, polymer **264** may build up on surfaces of the mask **261** and exposed surfaces of the etchable material **260** (i.e. sidewalls and bottom of the recess **262**). The polymer **264** may serve to advantageously passivate the mask **261** and sidewalls of the recess **262** by forming a mask surface passivation layer **274** and a sidewall passivation layer **275** respectively. For example, the polymer **264** may be a carbon polymer. Due to geometrical (and other) considerations, the polymer **264** may build up in larger quantities at the surface of the mask **261** than at the sidewalls (as shown).

(69) At the bottom of the recess **262**, a mixing layer **265** may also form. For example, the mixing layer may include various compounds formed from elements originating from both the etchable material **260** and the etchant. In some embodiments, the mixing layer **265** comprises both Si and O and in one embodiment, the mixing layer **265** is formed primarily through interactions of CF_2 and SiO_2 species. The mixing layer **265** may advantageously be particularly susceptible to removal via bombardment of energetic ions (e.g. CF_2^{+} ions).

(70) Several desirable characteristics are illustrated schematically in FIG. 2. For example, the recess **262** exhibits vertical sidewalls all the way to the bottom of the recess **262**. Specifically, widening of the recess **262** is not experienced at the top, in the middle (e.g. no bowing), or at the bottom of the recess **262**. Additionally, the polymer **264** is desirably formed on the mask **261** and in the recess **262**. For example, the mask surface passivation layer **274** does not extend over the

opening of the recess **262**, but also does not expose any portion of the mask **261**. Further, the sidewall passivation layer **275** is formed uniformly to the bottom of the recess **262** and is thinner than the mask surface passivation layer **274**. A mixing layer **265** is formed at the bottom (and may extend a small amount up the sides as shown in FIG. **1C**) of the recess **262**.

(71) The desirable characteristics shown in FIG. **2** may be achieved using the embodiment plasma processing methods described herein. In particular, independent (or nearly independent) control of the flux of the FC radicals **71**, the F radicals **72**, and the ions **73** at various the plasma process may advantageously facilitate the desirable characteristics. For example, providing sufficient FC radicals **71** may provide the depicted mask protection and mitigate sputtering. Further, sufficient FC radicals **71** may also serve to protect the sidewalls.

(72) Additionally, providing sufficient F radicals **72** and ions **73** may prevent clogging of the opening to the recess **262**. That is, the F radicals **72** may react with the polymer **264** and the ions **73** may dislodge the polymer **264** via bombardment to provide thickness control of the mask surface passivation layer **274** (and also the sidewall passivation layer **275**). Further, sufficient vertical ions **73** may be provided to prevent bowing of the recess **262** (e.g. due to etching of the sidewalls from wide angle ions). Similarly, providing sufficient energetic ions **73** and FC radicals **71** may advantageously maintain the etching process (e.g. at the mixing layer **265**).

(73) Other factors may also affect the control of the delivery of the FC radicals **71**, the F radicals **72**, and the ions **73** to the substrate. For example, pressure in the plasma processing chamber may affect ion energy and ion flux. Additionally, the degree that the radical flux can be modulated may depend on the residence time.

(74) All of the above considerations may be important to maintain a consistent etch profile and a constant etch rate over time. The desired (i.e. optimal) flux of the various species may differ depending on the phase in a cycle of the plasma process. Moreover, the system may change over time. Consequently, the control of the parameters may be dynamically changed in subsequent cycles of the plasma process to advantageously maintain a constant etch rate and consistent etch profile (e.g. tens of minutes into an etching process).

(75) For example, initially an SP pulse and a BP pulse may be utilize as the etch rate is higher when the recess is more shallow. However, as the process continues and the recess becomes deep the etch rate may slow. The BP pulse may be lengthened and/or a BP power spike (e.g. a corner etch phase) may be utilized to compensate and maintain a consistent etch profile and constant etch rate. Dynamic control of the plasma parameters may play a major role when the plasma process is not self-limited.

(76) FIG. **3** illustrates a schematic diagram of another target substrate during an example plasma process utilizing imbalanced control of plasma parameters in accordance with an embodiment of the invention. Similarly labeled elements may be as previously described.

(77) Referring to FIG. **3**, a target substrate **304** includes an etchable material **360**, a mask **361**, a recess **362**, and a polymer **364** disposed over the mask **361**. The depiction of the target substrate **304** is similar to that of the target substrate **204** except that various undesirable effects from providing excessive or insufficient amounts of the FC radicals **71**, F radicals **72**, and ions **73** at the recess **362** are demonstrated rather than the desirable effects of FIG. **2**. The sidewall passivation layer is not shown for clarity.

(78) As previously discussed, sufficient quantities of FC radicals **71** (FC radical flux $\Gamma_{\text{sub.FC}}$) may have desirable effects during a plasma process. However, an excessive FC radical flux **74** at the recess **262** may result in an overhang that undesirably prevents or reduces (clogs) further etching. In contrast, an insufficient FC radical flux **75** may result in too little polymer **364** buildup in the mask surface and undesirably widen the opening of the recess **362** as the mask edge is eroded. Such widening may also cause the sidewalls to be undesirably sloped (“V” shaped). Erosion may result in critical dimension (CD) loss (e.g. tapering).

(79) Similarly, sufficiently energetic ions (ion energy $\epsilon_{\text{sub.i}}$) may also have desirable effects. Yet

excessive ion energy **76** may remove polymer **364** protecting the mask and facilitate undesirable mask erosion. Conversely, insufficient ion energy **77** may allow polymer **364** to remain at the opening of the recess **362** and cause clogging. The low ion energy may also prevent ions from reaching the bottom of the recess **362** and increase the angular distribution of the ions causing shallow etch depth (e.g. reducing sidewall verticality) and bowing (e.g. due to lack of sidewall passivation) as shown. The ions **73** may also lose energy due to sidewall scatter further lowering the etch rate.

(80) For this reason providing BP during an SP phase (e.g. the first on phase **110** of FIG. **1A**) may be advantageously afford thickness control for polymer buildup over time. Preventing the clogging of the opening of the recess **362** may be particularly important when the mask **361** is thick. For example, a NAND mask may be 2 μm thick. However, even hard masks used in HARC processes may benefit from optimal polymer deposition to ensure high aspect ratios (e.g. 100 or more). Further, selectivity may rely on polymer deposition, but too much reduces or prevents etching.

(81) Sufficient F radicals **72** (F radical flux $\Gamma_{\text{sub.F}}$) also provide beneficial effects within an optimal range. But, as with the FC radicals **71** and the ions **73**, too little or too much can be problematic. For example, excessive F radical flux **78** may lead to erosion while insufficient F radical flux **79** may lead to clogging.

(82) FIGS. **4A**, **4B**, and **4C** illustrate another example plasma processing method in accordance with an embodiment of the invention, where FIG. **4A** illustrates a schematic timing diagram of the plasma processing method, FIG. **4B** illustrates a corresponding qualitative graph, and FIG. **4C** illustrates a corresponding target substrate. The plasma processing method of FIGS. **4A**, **4B**, and **4C** may be combined with or be a specific implementation of other plasma processing methods described herein such as the plasma processing method of FIGS. **1A**, **1B**, and **1C**, for example. Similarly labeled elements may be as previously described.

(83) Referring to FIGS. **4A** and **4B**, a schematic timing diagram **400** includes source power SP and bias power BP provided as pulses to generate reactive species and energetic ions over various phases of a cycle **450**. The cycle **450** includes various phases characterized by power and duration of both the SP and the BP. The cycle **450** includes a first on phase **410** during which an SP pulse **412** is applied with power $P_{\text{sub.S1}} > 0$ and a second on phase **420** during which a BP pulse **424** is applied with power $P_{\text{sub.B2}} > 0$.

(84) In addition to the first on phase **410** and the second on phase **420** (which may be similar to the first on phase **110** and the second on phase **120** of FIG. **1A**), the schematic timing diagram **400** also includes a first off phase **415** preceding the beginning of the BP pulse **424** as well as a second off phase **425** following the end of the BP pulse **424**. The first off phase **415** is a delay between the SP phase and the BP phase of the plasma process.

(85) The second off phase **425** is also a delay, but between the BP phase and a subsequent SP phase. For example, the second off phase **425** may also serve as a by-product management phase (e.g. be similar to the optional fourth phase **140** of FIG. **1A**). Alternatively, the second off phase **425** may be an independent phase included in addition to or instead of a by-product management phase.

(86) A qualitative graph **402** including an ion flux curve **405** corresponding to the schematic timing diagram **400** is illustrated in FIG. **4B**. The behavior of the ion flux $\Gamma_{\text{sub.i}}$ in the first on phase **410** and the second on phase **420** is similar to the analogous phases in FIG. **1B**. However, in contrast, $\Gamma_{\text{sub.i}}$ decreases substantially to zero during the first off phase **415** and the second off phase **425**. This effective pause in ion delivery allows chemistry control and may advantageously facilitate desirable polymer buildup control.

(87) Referring now to FIG. **4C**, a target substrate **404** includes an etchable material **406**, a mask **461**, and polymer **464** formed on the mask **461** and in a recess **462**. As shown, the delays between the SP phase and the BP phase may afford chemistry control decoupled from ion control by allowing the ratio of $F:CF_{\text{sub.2}}$ to decrease before applying the BP pulse. The increased

concentration of C at the target substrate **404** may advantageously allow polymer **464** to build up thereby protecting the mask **461** and sidewalls of the recess **462** from ion bombardment damage. (88) Since the polymer buildup is a neutral-driven process (indicated by FC radicals **71**), polymer **464** may primarily form on the mask surface rather than at the bottom of the recess **462**. In the second on phase **420**, ions **73** are delivered to the bottom of the recess **462** perform the main etch shown by cycle etch depth **466**. The mask **461** may be advantageously protected by the polymer **464** built up during the first off phase **415**. Similarly, the F radicals **72** generated in the first on phase **410** may also remove polymer **464**. The second off phase **425** may provide the benefit of allowing polymer **464** to build up and protect the mask **461** during the first on phase **410**.

(89) The first off phase duration $t_{\text{sub.O1}}$ (of the first off phase **415**) and the second off phase duration $t_{\text{sub.O2}}$ (of the second off phase **425**) may be appropriately chosen to allow sufficient polymer to build up and protect the mask **461** without building up too much polymer which could result in clogging. In various embodiments, $t_{\text{sub.O1}}$ is greater than about 5 μs . In some embodiments, $t_{\text{sub.O1}}$ is between about 5 μs and about 50 μs . However in other embodiments, $t_{\text{sub.O1}}$ is between about 5 μs and about 10 μs . In one embodiment, the first off phase duration $t_{\text{sub.O1}}$ is about 20 μs .

(90) The second off phase duration $t_{\text{sub.O2}}$ may be similar to or different from the first off phase duration $t_{\text{sub.O1}}$. In general, $t_{\text{sub.O2}}$ may be shorter than $t_{\text{sub.O1}}$ (e.g. due to the reduced rate of polymer erosion in the first on phase **410** as opposed to the second on phase **420**). In various embodiments $t_{\text{sub.O2}}$ is greater than about 5 μs . In some embodiments, $t_{\text{sub.O2}}$ is between about 5 μs and about 10 μs . In one embodiment, the second off phase duration $t_{\text{sub.O2}}$ is about 5 μs .

(91) FIG. 5 illustrates a schematic timing diagram of yet another example plasma processing method in accordance with an embodiment of the invention. The plasma processing method of FIG. 5 may be combined with or be a specific implementation of other plasma processing methods described herein such as the plasma processing method of FIGS. 1A, 1B, and 1C, for example. Similarly labeled elements may be as previously described.

(92) Referring to FIG. 5, a schematic timing diagram **500** includes source power SP and bias power BP provided as pulses to generate reactive species and energetic ions over various phases of a cycle **550**. The cycle **550** includes various phases characterized by power and duration of both the SP and the BP. The cycle **550** includes a first on phase **510** during which a first SP pulse **512** is applied with power $P_{\text{sub.S1}} > 0$ and a second on phase **520** during which a BP pulse **524** is applied with power $P_{\text{sub.B2}} > 0$ in addition to a second SP pulse **522** with power $P_{\text{sub.S2}} < P_{\text{sub.S1}}$.

(93) A first off phase **515** and a second off phase **525** are also included during which no BP is provided. As shown, the second SP pulse **522** may be provided during one or both of the first off phase **515** and the second off phase **525**. For example, the SP may be maintained at a lower level until the following SP phase. The lower SP during the second on phase **520** in combination with the first off phase **515** may advantageously enable further chemistry control and/or ion control.

(94) FIG. 6 illustrates a schematic timing diagram of still another example plasma processing method in accordance with an embodiment of the invention. The plasma processing method of FIG. 6 may be combined with or be a specific implementation of other plasma processing methods described herein such as the plasma processing method of FIGS. 1A, 1B, and 1C, for example. Similarly labeled elements may be as previously described.

(95) Referring to FIG. 6, a schematic timing diagram **600** includes source power SP and bias power BP provided as pulses to generate reactive species and energetic ions over various phases of a cycle **650** which includes a first on phase **610** during which a first SP pulse **612** is applied with power $P_{\text{sub.S1}} > 0$, a first off phase **615** during which a second SP pulse **622** is applied with power $P_{\text{sub.SO1}} < P_{\text{sub.S1}}$, a second on phase **620** during which a BP pulse **624** is applied with power $P_{\text{sub.B2}} > 0$, and an optional second off phase **625** during which no SP or BP is applied.

(96) In contrast to the plasma processing method of FIG. 5, the SP is not maintained in the second on phase **520** or the optional second off phase **525**. Optionally, the second SP pulse **622** may be

extended into (or even through) the second on phase **620** as illustrated by the optional second SP pulse extension **623**. The additional current during the first off phase **615** but not in the optional second off phase **625** may advantageously facilitate increased polymerization during the first off phase **615** compared to the second off phase **625**.

(97) FIGS. **7A**, **7B**, and **7C** illustrate still yet another example plasma processing method in accordance with an embodiment of the invention, where FIG. **7A** illustrates a schematic timing diagram of the plasma processing method, FIG. **7B** illustrates a corresponding qualitative graph, and FIG. **7C** illustrates a corresponding target substrate. The plasma processing method of FIGS. **7A**, **7B**, and **7C** may be combined with or be a specific implementation of other plasma processing methods described herein such as the plasma processing method of FIGS. **1A**, **1B**, and **1C**, for example. Similarly labeled elements may be as previously described.

(98) Referring to FIG. **7A**, a schematic timing diagram **700** includes source power SP and bias power BP provided as pulses to generate reactive species and energetic ions over various phases of a cycle **750** which includes a first on phase **710** during which a first SP pulse **712** is applied with power $P_{S1} > 0$, a first off phase **715**, a second on phase **720** during which BP is applied, and an optional second off phase **725**. The duration of the second on phase **720** may be on the order of microseconds, such as 20 μ s.

(99) In contrast to the second on phase **420** of FIG. **4A**, the second on phase **720** includes a BP pulse train **726** comprising a plurality of BP spikes **728**. For example, each BP spike **728** may be a DC pulse of short duration (e.g. 1 μ s DC pulses, 400 kHz pulses, or longer such as 13 MHz, 1 MHz, 900 kHz, etc.). Alternatively, each BP spike **728** may comprise a train of DC pulses.

(100) Referring now to FIGS. **7B** and **7C**, a qualitative graph **702** and a qualitative diagram of the ion energy and angle distribution **703** (IEADF) corresponding to the schematic timing diagram **700** is shown. The qualitative graph **702** includes an ion flux curve **705** similar to ion flux curve **405** of FIG. **4B** except that $\Gamma_{sub,i}$ drops to substantially zero after each BP spike **728** during the second on phase **720**.

(101) As shown in block **751**, the ions have medium energy and wide angle during the first on phase **710**. During the first off phase **715**, the ions still have a wide angle, but much less energy (e.g. thermal) as shown in block **752**. Each BP spike **728** of the BP pulse train **726** during the second on phase **720** generates highly energetic ions with a narrow angular distribution as shown in block **753**. Block **754** again illustrates the very low ion energy with the SP and BP are off during the second off phase **725**.

(102) The inclusion of the BP pulse train **726** in the second on phase **720** may advantageously reduce the flux of ions with wide angular distribution during the BP phase. For example, ions may lose verticality while traveling through well-developed sheaths (e.g. due to collisions). The sheath may become more than more developed over time with applied BP. Providing short duration BP spikes **728** may prevent the sheath from becoming well developed and advantageously deliver high verticality ions throughout the second on phase **720**.

(103) FIG. **8** illustrates a block diagram of an example CCP system in accordance with an embodiment of the invention. The CCP system of FIG. **8** may be used to implement the schematic timing diagrams to perform any of the embodiment methods as described herein, such as the schematic timing diagram of FIG. **1A**, for example. Furthermore, the CCP system of FIG. **8** may be used to perform any of the embodiment methods as described herein, such as the methods of FIGS. **9-11**, for example. Similarly labeled elements may be as previously described.

(104) Referring to FIG. **8**, a CCP system **801** includes a target substrate **804** disposed in a plasma processing chamber **884** inside which a CCP plasma **886** is generated between an SP electrode **888** and a BP electrode **889**. The target substrate **804** may be electrically coupled to the BP electrode **889**. For example, the BP electrode **889** may be a substrate holder and is an electrostatic chuck in one embodiment.

(105) SP is provided to the SP electrode **888** via an SP control path **887** while BP is provided to the

BP electrode **889** via a BP control path **897**. SP provided to the SP electrode **888** generates the CCP plasma **886** within the plasma processing chamber **884**. The SP control path **887** includes an SP function generator **885** coupled to the SP electrode **888** through an SP impedance matching network **883**. The SP function generator **885** is coupled to an SP pulse modulation circuit **881** which outputs a modulated source signal. The SP function generator **885** superimposes a waveform (e.g. a sinusoidal waveform) onto the modulated source signal to generate SP pulses that are delivered to the SP electrode **888**.

(106) Similarly, the BP control path **897** includes an optional BP function generator **895** coupled to the BP electrode **889** through an optional BP impedance matching circuit **893**. The BP control path **897** also includes an optional BP pulse modulation circuit **891**. The optional BP function generator **895** may be coupled to the optional BP pulse modulation circuit **891** which outputs a modulated source signal. The optional BP function generator **895** may superimpose a waveform (e.g. a sinusoidal waveform) onto the modulated source signal to generate BP pulses that are delivered to the BP electrode **889**. Both the SP function generator **885** and the optional BP function generator **895** may include an amplification circuit to allow adjustment of amplitude of SP pulses and BP pulses during different phases of a pulse cycle.

(107) Alternatively, the optional BP function generator **895** and the optional BP impedance matching circuit **893** may be omitted and BP may be directly supplied to the BP electrode **889**. Such a configuration may be used with DC power is provided directly to the BP electrode **889**. For example, a BP pulse train may be generated at the BP electrode **889** by coupling the optional BP pulse modulation circuit **891** directly to the BP electrode **889**.

(108) In the specific case where the BP pulses are identical to the SP pulses except for a timing offset the BP control path **897** may use the SP pulse modulation circuit **881** and omit the optional BP pulse modulation circuit **891**. However, the optional BP pulse modulation circuit **891** may advantageously enable greater flexibility regarding the pulse width, amplitude, and shape of the BP pulses compared to the SP pulses.

(109) On or more elements described above may be included in controller **880** that controls the timing of the SP pulses and the BP pulses. A pulse modulation timing circuit **882** is also included in the controller **880**. The pulse modulation timing circuit **882** may accept an input from the SP pulse modulation circuit **881** and adjust the timing of the signal modulated by the optional BP pulse modulation circuit **891**. The pulse modulation timing circuit **882** may also control the timing of both the SP pulse modulation circuit **881** and the optional BP pulse modulation circuit **891**.

(110) Although the CCP system **801** is illustrated as providing SP at a top electrode and BP at a bottom electrode, this does not have to be the case. For example, in an alternate configuration, the CCP system **801** may provide both SP and BP at the bottom electrode while the top electrode operates as a counter electrode.

(111) FIG. **9** illustrates an example method of plasma processing in accordance with an embodiment of the invention. The method of FIG. **9** may be combined with other methods and performed using the systems and apparatuses as described herein. For example, the method of FIG. **9** may be combined with any of the embodiments of FIGS. **1A-8**. Although shown in a logical order, the arrangement and numbering of the steps of FIG. **9** are not intended to be limited. The method steps of FIG. **9** may be performed in any suitable order or concurrently with one another as may be apparent to a person of skill in the art.

(112) Referring to FIG. **9**, step **910** of a method **900** of plasma processing includes performing a first on phase comprising applying a SP pulse to an SP electrode to generate plasma in a plasma processing chamber, the SP pulse comprising a first SP power level and terminating at the end of the first on phase. Step **920** includes performing a second on phase after the first on phase, the second on phase comprising applying a first BP pulse to a BP electrode coupled to a target substrate within the plasma processing chamber, the first BP pulse comprising a first BP power level and accelerating ions of the plasma toward to target substrate to etch a recess in an etchable material of

the target substrate.

(113) A corner etch phase is performed after the second on phase in step **930**, the corner etch phase comprising applying a BP spike comprising a second BP power level greater than the first BP power level. The duration of the BP spike is less than the duration of the first BP pulse. Step **940** includes performing a by-product management phase after the corner etch phase, the by-product management phase comprising applying source power to the SP electrode at a second SP power level that is less than the first SP power level, and applying bias power to the BP electrode at a third BP power level that is less than the first BP power level. Steps **910**, **920**, **930**, and **940** may be performed cyclically as part of a cycle **950**.

(114) FIG. **10** illustrates another example method of plasma processing in accordance with an embodiment of the invention. The method of FIG. **10** may be combined with other methods and performed using the systems and apparatuses as described herein. For example, the method of FIG. **10** may be combined with any of the embodiments of FIGS. **1A-8**. Additionally, the method of FIG. **10** may be combined with other methods such as the method of FIG. **9**, for example. Although shown in a logical order, the arrangement and numbering of the steps of FIG. **10** are not intended to be limited. The method steps of FIG. **10** may be performed in any suitable order or concurrently with one another as may be apparent to a person of skill in the art.

(115) Referring to FIG. **10**, step **1010** of a method **1000** includes performing a glow phase comprising providing a first SP pulse comprising a first SP power level to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber, the first SP pulse terminating at the end of the glow phase, and providing a first BP pulse comprising a first BP power level to a BP electrode coupled to a target substrate within the plasma processing chamber. Step **1020** includes performing an afterglow phase after the glow phase, the afterglow phase comprising providing a second BP pulse to the BP electrode in an afterglow of the capacitively coupled plasma, the second BP pulse comprising a second BP power level that is less than the first BP power level. Steps **1010** and **1020** may be performed cyclically as part of a cycle **1050**.

(116) Example embodiments of the invention are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.

(117) Example 1. A method of plasma etching, the method including: performing a first on phase including applying a SP pulse to an SP electrode to generate plasma in a plasma processing chamber, the SP pulse including a first SP power level and terminating at the end of the first on phase; performing a second on phase after the first on phase, the second on phase including applying a first BP pulse to a BP electrode coupled to a target substrate within the plasma processing chamber, the first BP pulse including a first BP power level and accelerating ions of the plasma toward to target substrate to etch a recess in an etchable material of the target substrate; performing a corner etch phase after the second on phase, the corner etch phase including applying a BP spike including a second BP power level greater than the first BP power level, where the duration of the BP spike is less than the duration of the first BP pulse; and performing a by-product management phase after the corner etch phase, the by-product management phase including applying source power to the SP electrode at a second SP power level that is less than the first SP power level, and applying bias power to the BP electrode at a third BP power level that is less than the first BP power level.

(118) Example 2. The method of example 1, further including: cyclically performing the first on phase, the second on phase, the corner etch phase, and the by-product management phase.

(119) Example 3. The method of one of examples 1 and 2, where the first on phase further includes: applying a second BP pulse concurrently with the SP pulse, the second BP pulse including a fourth BP power level greater than the first BP power level.

(120) Example 4. The method of example 3, where applying the second BP pulse includes decreasing the fourth BP power level to the first BP power level prior to the termination of the SP pulse.

(121) Example 5. The method of one of examples 1 to 4, further including: performing an off phase after the first on phase and before the second on phase, the off phase including applying no bias power to the BP electrode.

(122) Example 6. The method of one of examples 1 to 5, where both the second SP power level and the third BP power level are greater than zero.

(123) Example 7. A method of plasma processing, the method including: performing a glow phase including providing a first SP pulse including a first SP power level to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber, the first SP pulse terminating at the end of the glow phase, and providing a first BP pulse including a first BP power level to a BP electrode coupled to a target substrate within the plasma processing chamber; and performing an afterglow phase after the glow phase, the afterglow phase including providing a second BP pulse to the BP electrode in an afterglow of the capacitively coupled plasma, the second BP pulse including a second BP power level that is less than the first BP power level.

(124) Example 8. The method of example 7, further including: cyclically performing the glow phase and the afterglow phase.

(125) Example 9. The method of one of examples 7 and 8, where the SP includes high frequency power greater than about 27 MHz, and where the BP includes low frequency power between about 100 kHz and about 13 MHz.

(126) Example 10. The method of one of examples 7 to 9, where the afterglow phase further includes: providing a second SP pulse to the SP electrode, the second SP pulse including a second SP power level that is less than the first SP power level.

(127) Example 11. The method of one of examples 7 to 10, further including: performing a first off phase after the glow phase, the first off phase including applying no bias power to the BP electrode to delay providing the second BP pulse.

(128) Example 12. The method of example 11, where the first off phase further includes applying a second SP pulse to the SP electrode, the second SP pulse including a second SP power level that is less than the first SP power level.

(129) Example 13. The method of one of examples 11 and 12, further including: performing a second off phase after the afterglow phase, the second off phase including applying to no bias power to the BP electrode to delay a subsequent glow phase.

(130) Example 14. The method of example 13, where the first off phase, the afterglow phase, and the second off phase further include applying a second SP pulse to the SP electrode, the second SP pulse including a second SP power level that is less than the first SP power level.

(131) Example 15. The method of one of examples 7 to 14, where the afterglow phase further includes providing a BP spike to the BP electrode after the second BP pulse, the BP spike including a third BP power level greater than the second BP power level, where the duration of the BP spike is less than the duration of the second BP pulse.

(132) Example 16. The method of one of examples 7 to 15, where the second BP pulse is a BP pulse train including a plurality of BP spikes, where the BP is direct current (DC) power.

(133) Example 17. A method of plasma processing, the method including cyclically performing the following steps: concurrently applying SP including a first SP power level to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber, and BP including a first BP power level to a BP electrode coupled to a target substrate within the plasma processing chamber; reducing the SP from the first SP power level to a second SP power level; reducing the BP from the first BP power level to a second BP power level; and reducing the BP from the second BP power level to a third BP power level.

(134) Example 18. The method of example 17, further including: decreasing the BP from the first BP power level to the second BP power level before reducing the SP from the first SP power level to the second SP power level.

(135) Example 19. The method of one of examples 17 and 18, where reducing the BP from the first

BP power level to the second BP power level includes: reducing the BP from the first BP power level to zero; applying no bias power to the BP electrode for a nonzero duration; and increasing the BP from zero to the second BP power level.

(136) Example 20. The method of one of examples 17 to 19, further including: after reducing the BP from the first BP power level the second BP power level and before reducing the BP from the second BP power level to the third BP power level, increasing the BP from the second BP power level to a fourth BP power level; and decreasing the BP from the fourth BP power level to the third BP power level.

(137) While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of plasma processing, the method comprising cyclically performing a cycle comprising the following steps: performing a glow phase comprising providing a first source power (SP) pulse comprising a first SP power level for a first duration to an SP electrode to generate a capacitively coupled plasma in a plasma processing chamber, the first SP pulse terminating at the end of the glow phase; and performing an afterglow phase after the glow phase, the afterglow phase comprising providing a bias power (BP) pulse train to a BP electrode coupled to a target substrate within the plasma processing chamber in an afterglow of the capacitively coupled plasma for a second duration between about 10 μ s and about 100 μ s, the BP pulse train comprising a plurality of BP spikes, each of the plurality of BP spikes being a direct current (DC) pulse having a first BP power level.
2. The method of claim 1, wherein the glow phase further comprises providing a BP pulse comprising a second BP power level to the BP electrode, the second BP power level being greater than the first BP power level.
3. The method of claim 1, wherein each of the plurality of BP spikes has a short duration less than about 2.5 μ s.
4. The method of claim 1, wherein the plurality of BP spikes of the BP pulse train are provided to the BP electrode at a frequency greater than about 400 kHz.
5. The method of claim 1, wherein the afterglow phase further comprises: providing a second SP pulse to the SP electrode, the second SP pulse comprising a second SP power level that is less than the first SP power level.
6. The method of claim 1, wherein the cycle further comprises: performing a first off phase after the glow phase and before the afterglow phase, the first off phase comprising providing no source power to the SP electrode and no bias power to the BP electrode during the first off phase.
7. The method of claim 6, wherein each of the plurality of BP spikes has a short duration less than about 2.5 μ s.
8. The method of claim 1, wherein the afterglow phase further comprises providing a single BP spike to the BP electrode after the BP pulse train, the single BP spike comprising a BP spike power level greater than the first BP power level, wherein the duration of the single BP spike is less than the duration of the BP pulse train.
9. A method of plasma etching, the method comprising: performing a first on phase comprising applying a source power (SP) pulse to an SP electrode to generate plasma in a plasma processing chamber, the SP pulse comprising a first SP power level during the first on phase; performing a second on phase after the first on phase, the second on phase comprising applying a bias power (BP) pulse train to a BP electrode coupled to a target substrate within the plasma processing

chamber, the BP pulse train comprising a plurality of BP spikes, each of the plurality of BP spikes being a direct current (DC) pulse having a first BP power level and accelerating ions of the plasma toward the target substrate to etch a recess in an etchable material of the target substrate; performing a corner etch phase after the second on phase, the corner etch phase comprising applying a single BP spike comprising a second BP power level greater than the first BP power level, wherein the duration of the single BP spike is less than the duration of the BP pulse train; and performing a by-product management phase after the corner etch phase, the by-product management phase comprising applying source power to the SP electrode at a second SP power level that is less than the first SP power level, and applying bias power to the BP electrode at a third BP power level that is less than the first BP power level.

10. The method of claim 9, wherein each of the plurality of BP spikes has a short duration less than about 2.5 μ s.

11. The method of claim 9, wherein the plurality of BP spikes of the BP pulse train are provided to the BP electrode at a frequency greater than about 400 kHz.

12. The method of claim 9, further comprising: cyclically performing a cycle comprising the first on phase, the second on phase, the corner etch phase, and the by-product management phase.

13. The method of claim 9, wherein the first on phase further comprises: applying a BP pulse concurrently with the SP pulse, the BP pulse comprising a fourth BP power level greater than the first BP power level.

14. The method of claim 9, wherein both the second SP power level and the third BP power level are greater than zero.

15. A method of plasma etching, the method comprising: performing a first on phase comprising applying a source power (SP) pulse to an SP electrode to generate plasma in a plasma processing chamber, the SP pulse comprising a first SP power level during the first on phase; performing a first off phase with a duration greater than about 5 μ s after the first on phase, the first off phase comprising reducing the first SP power level to zero and applying no bias power; performing a second on phase after the first off phase, the second on phase comprising applying a bias power (BP) pulse train to a BP electrode coupled to a target substrate within the plasma processing chamber, the BP pulse train comprising a plurality of BP spikes, each of the plurality of BP spikes being a direct current (DC) pulse having a first BP power level and accelerating ions of the plasma toward the target substrate to etch a recess in an etchable material of the target substrate; and performing a corner etch phase after the second on phase, the corner etch phase comprising applying a single BP spike comprising a second BP power level greater than the first BP power level, wherein the duration of the single BP spike is less than the duration of the BP pulse train.

16. The method of claim 15, wherein the plurality of BP spikes of the BP pulse train are provided to the BP electrode at a frequency greater than about 400 kHz.

17. The method of claim 15, further comprising: performing a by-product management phase after the corner etch phase, the by-product management phase comprising applying source power to the SP electrode at a second SP power level that is less than the first SP power level, and applying bias power to the BP electrode at a third BP power level that is less than the first BP power level.

18. The method of claim 17, wherein both the second SP power level and the third BP power level are greater than zero.

19. The method of claim 15, further comprising: cyclically performing a cycle comprising the first on phase, the first off phase, the second on phase, and the corner etch phase.

20. The method of claim 15, wherein the first on phase further comprises: applying a BP pulse concurrently with the SP pulse, the BP pulse comprising a BP power level greater than the first BP power level.
